

CSBS SEMESTER 4 SYLLABUS

Subjects, Modules, Topics & Subtopics

DISCRETE MATHEMATICS

Course Code: PCCCS401

Module 1: Sets, Relation and Function, Principles of mathematical induction

Subtopics: Operations and Laws of Sets, Cartesian Products, Binary Relation, Partial Ordering Relation, Equivalence Relation, Image of a Set, Sum and Product of Functions, Bijective functions, Inverse and Composite Function, Size of a Set, The Well-Ordering Principle, The Division algorithm, Prime Numbers, The Greatest Common Divisor, Euclidean Algorithm, The Fundamental Theorem of Arithmetic, Coprimality (Euler's totient function), Chinese Remainder Theorem

Module 2: Basic counting techniques, Propositional Logic, Proof Techniques

Subtopics: Inclusion and exclusion principle, pigeon-hole principle, permutation and combination, recurrence relations, generating functions, Propositional Logic (Syntax, Semantics, Validity and Satisfiability, Basic Connectives and Truth Tables, Logical Equivalence, Laws of Logic, Logical Implication, Rules of Inference, Quantifiers), Proof Techniques (Proof Methods and Strategies, Forward Proof, Proof by Contradiction, Proof by Contraposition), Boolean Algebra (Identities, Duality, Representation of Boolean Function, Disjunctive and Conjunctive Normal Form)

Module 3: Algebraic Structures and Morphism

Subtopics: Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups, Permutation Groups, Normal Subgroups, Ring, Field, Vector spaces, Inner-product spaces

Module 4: Graphs and Trees

Subtopics: Graphs and their properties, Degree, Connectivity, Path, Cycle, Sub Graph, Isomorphism, Eulerian and Hamiltonian Walks, Graph Coloring, Planar Graphs, Matching, Trees

COMPUTER ORGANIZATION & ARCHITECTURE

Course Code: PCCCS402

Module 1: Introduction and Performance Evaluation

Subtopics: Role of abstraction, basic functional units of a computer, Stored Program Architecture, VonNeumann model of computation, Moore's law, clocking methodology, Amdahl's law, Notion of IPC, performance

Module 2: Data representation and basic operations

Subtopics: Fixed and floating point (IEEE 754 Single and double precision format) representation of numbers, Overflow, Design of Adders (Ripple Carry Adder, Carry Look Ahead Adder), multiplication (shift-and-add, Booth multiplier, carry save multiplier), Division (non-restoring and restoring techniques), floating point arithmetic

Module 3: Instruction Set Architecture

Subtopics: CPU registers, instruction format and encoding, addressing modes, instruction set, instruction types, instruction decoding and execution, basic instruction cycle, RISC, CISC, Case study of common CPUs

Module 4: Processor Design

Subtopics: Hardwired and micro-programmed design approaches, Case study of simple hypothetical CPU design

Module 5: Memory hierarchy

Subtopics: Memory hierarchy, Main memory organization (paging, segmentation, virtual memory), Cache memory (different indexing mechanisms, Trade-offs related to block size, associativity, and cache size), Processor-cache interactions, write through/write-back caches, Average memory access time, Cache replacement policies (LRU), locality of reference, Memory interleaving, magnetic disks

Module 6: Input/Output Organization

Subtopics: Programmed I/O, Interrupt-driven I/O, DMA

Module 7: Parallel Processing

Subtopics: Pipelining (Basic concepts, instruction and arithmetic pipeline, dependencies and hazards, techniques for handling hazards, Pipeline optimization), Superscalar, super pipelined and VLIW processor architectures, Array and vector processors, Multiprocessor architecture (taxonomy of parallel architectures, Centralized shared-memory architecture, distributed shared memory architecture), Cache Coherence

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Course Code: PCCCS403

Module 1: Introduction to Artificial Intelligence

Subtopics: Definition and Scope of AI, History & Evolution of AI, Sub fields of AI, Ethical Considerations, Problem Solving in AI (Searching, Optimization, Planning, Game Theory), Knowledge Representation (Propositional Logic, Predicate Logic)

Module 2: Problem solving techniques

Subtopics: State space search, control strategies, heuristic search, problem characteristics, production system characteristics, Generate and test, Hill climbing, best first search, A search, Constraint satisfaction problem, Mean-end analysis, Min-Max Search, Alpha-Beta Pruning, Iterative Deepening*

Module 3: Introduction to Machine Learning

Subtopics: Machine Learning Definition & Scope, Evolution of Learning Systems, Data Representation (Features, Covariance, Orthogonality, Feature Selection vs Feature Extraction, Missing Data, Outliers, Noise, Normalization, Standardization), Training Validation and Testing, Model Selection (Holdout, Cross validation, k-fold Cross validation, random sampling, Bias-Variance tradeoff, overfitting, underfitting), Machine Learning Types (Supervised, Unsupervised, Reinforcement, Adversarial, Meta-Learning), Transfer Learning, Ensemble Learning (Bagging and Boosting)

Module 4: Supervised Learning

Subtopics: Regression (Linear Regression, Polynomial Regression, regularization), Classification (Logistic Regression, Decision Trees, Bayesian Classifier, Parametric and Nonparametric estimation, Naïve Bayes Classifier, K-Nearest Neighbours, Support Vector Machines), Representation Learning (Perceptron, activation functions, XOR problem, Multilayered Perceptron, Gradient Descent, Backpropagation), Performance Metrics (Confusion Matrix, Precision, Recall, F-Measure, Area under ROC Curve)

Module 5: Unsupervised Learning

Subtopics: Dimensionality Reduction (Eigen Value Decomposition, Principal Component Analysis, Linear Discriminant Analysis, Singular Value Decomposition), Clustering (Gaussian Mixture Models, Expectation Maximization, K-Means Algorithms, Hierarchical Clustering), Metrics (Similarity Measures, silhouette coefficient, Cluster Linkage Metrics, Cluster Validity Indices)

Module 6: Reinforcement Learning

Subtopics: Definition and Key Concepts (Agent, Environment, Reward, Policy, Value Function), Algorithms (Q-Learning, Exploration vs Exploitation, Convergence, Issues and Challenges)

DESIGN & ANALYSIS OF ALGORITHMS

Course Code: PCCCS404

Module 1: Introduction

Subtopics: Characteristics of algorithm, Analysis of algorithm (Asymptotic analysis of complexity bounds - best, average and worst-case behavior), Performance measurements, Time and space trade-offs, Analysis of recursive algorithms (Substitution method, Recursion tree method, Master's theorem), Divide and Conquer algorithms (Merge Sort, Quick Sort, Finding lower bound, Strassen's algorithm)

Module 2: Fundamental Algorithmic Strategies

Subtopics: Bruteforce, Greedy, Dynamic Programming, Branch and Bound, Backtracking methodologies, Bin Packing, Knapsack, TSP, Heuristics and their application domains, KMP algorithm

Module 3: Graph and Tree Algorithms

Subtopics: Traversal algorithms (Depth First Search, Breadth First Search), Disjoint Set Data Structures, Shortest paths algorithms, Minimum Spanning Tree, Topological sorting, Network Flow Problem

Module 4: Tractable and Intractable Problems

Subtopics: Computability of Algorithms, Computability classes (P, NP, NP-complete and NP-hard), Cook's theorem, Standard NP-complete problems and Reduction techniques

Module 5: Advanced Topics

Subtopics: Approximation algorithms, Randomized algorithms, Class of problems beyond NP (P SPACE)

ADVANCED PROGRAMMING (OOP)

Course Code: PCCCS405

Module 1: Familiarity with the programming environment

Subtopics: Understanding the build system, IDE, debugging, profiling and source code management, Introduction to various programming paradigms, advantages of OOP, comparison of OOP with Procedural Paradigm

Module 2: Basic principles of the object-oriented development process

Subtopics: Introduction to Object Oriented Paradigm (Data encapsulation, modularity, code reuse, identifying classes, attributes, methods and objects, class relationships), Importing pre-written classes using the this keyword, calling and defining methods, writing and instantiating classes, setter/getter methods, instance variables, returning values

Module 3: Advanced features of OOP

Subtopics: Polymorphism, Inheritance, abstract classes & Interfaces, copying and cloning objects, Wrapper class, Stream, I/O operations, Parameter polymorphism, method resolution, partially and fully overriding methods, Comparable/Comparator interface, Cloneable interface

Module 4: Access modifiers, Exceptions & Multithreading

Subtopics: Access modifiers, packages, Java APIs, Exception handling (try/catch block, nesting try/catch blocks, throw and throws keywords), Java Thread Programming (Thread Synchronization and Thread Communication)

Module 5: Collection Framework

Subtopics: Language supported libraries for handling advanced data structures (Hierarchy of Collection Framework, ArrayList, LinkedList, Vector, HashSet), Time complexity analysis, sorting objects, iterating over objects

Module 6: Modeling and Design patterns

Subtopics: Basic modeling techniques (Class diagram, sequence diagram, use case diagrams), Introduction to design patterns (iterator, singleton, flyweight, adapter, strategy, template, prototype, factory, façade, decorator, composite, proxy, chain of responsibility, observer, state), Model View Controller

Module 7: Basic Android Programming & UI Design

Subtopics: Android Components (Activity, Services, Content Provider, Broadcast Receiver), Simple UI Design, Applet and Swing, Android app making, stand alone and web app development